KrishiSahayak

Predicting Distress Risk Before Harvest Using Rainfall, Temperature, andHistorical Yield Patterns Across Maharashtra Districts

A Machine Learning-Based Early Warning System for District-Level Agricultural Distress RiskClassification

Regression Analysis Assignment (Predictive Modelling)

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1. Introduction

Maharashtra is one of India's largest agricultural states, with a substantial share of its rural population directly dependent on rain-fed farming across crops such as cotton, soybean, jowar, andsugarcane. Agricultural output in the state is strongly shaped by monsoon rainfall variability, andyears of below-normal or erratic rainfall frequently translate into reduced yields and financial hardship for farming households. Conventionally, the true extent of a season's agricultural distressbecomes clear only after harvest, when yield and market outcomes are already known -- by whichpoint relief measures, crop insurance claims, and support interventions arrive too late to prevent thehardship they are meant to address.

The human cost of this delay is well documented. According to National Crime Records Bureau(NCRB) data, Maharashtra has recorded the highest number of farmer suicides among Indianstatesin recent years, accounting for approximately 38% of the national total. Independent researchexamining year-on-year data across several agricultural states, including Maharashtra, foundaconsistent pattern: suicide rates among farmers are measurably higher in years with a rainfall deficit. This documented linkage between weather-driven crop failure and farmer distress is central tothemotivation for this project -- an accessible, early-warning classification tool, built substantially onthesame rainfall-deviation signal this project's own analysis later identifies as the single strongest predictor of distress (Sections 6 and 9), could help give agricultural officers and support programsawindow for intervention that does not currently exist. Many farming households, particularlysmallholders in drought-prone regions, also lack easy access to detailed, localised informationabout how their specific district and crop combination is trending relative to historical norms -- a gapthisproject's dashboard is designed to help close.

KrishiSahayak addresses this gap using a supervised machine learning approach. Rather thanwaitingfor a season's outcome, the system uses early-season rainfall, temperature, sown area, andadistrict-crop combination's own recent yield history to classify likely distress risk -- Low, Medium, orHigh -- before the season concludes. The project combines a rigorous, leakage-free data pipelinebuilt from real Maharashtra government agricultural statistics with a comparison of a baselinelinearmodel (Logistic Regression) against a stronger, non-linear model (XGBoost), and is deployedas anaccessible, bilingual (English-Marathi) Streamlit dashboard intended for agricultural department officers and extension workers.

2. Problem Statement

Formally, the task addressed by this project is: given a district, a crop, and a year's early-seasonconditions -- rainfall, temperature, and area sown -- together with that specific district-cropcombination's own recent yield history, classify the combination into one of three distress riskcategories (Low, Medium, High) before the season's actual yield outcome is known.

A critical design constraint of this project is that the current year's actual yield is never usedasamodel input; it is used only to construct the target label during data preparation. This is a deliberatesafeguard against a common and serious mistake in predictive modelling -- using a disguisedversionof the outcome itself as a predictor, which produces misleadingly high accuracy without any genuineearly-warning value. An earlier iteration of this pipeline

exhibited exactly this issue, and correctingit was a core part of this project's methodology (see Section 6, Data Pre-processing).

The intended users of this system are agricultural department officers, insurance and relief planners, and extension workers who need an early, data-backed signal of which district-crop combinationsare heading toward distress -- not, at this stage, individual farmers directly, though the bilingual dashboard is designed simply enough to extend to that audience in future work.

3. Objectives

- Build a clean, leakage-free, model-ready dataset from real Maharashtra government cropstatistics merged with rainfall and temperature records.
- Engineer a composite distress score and a three-class target label without allowing any current-season outcome data to leak into the predictors.
- Train and rigorously compare a baseline model (Logistic Regression) against a stronger candidate model (XGBoost) on identical data splits.
- Validate model performance using both a stratified train-test split and stratified k-foldcross- validation, rather than relying on a single split.
- Interpret which factors most strongly drive predicted risk, rather than treating the model asan unexplainable black box.
- Deploy the resulting model as an accessible, bilingual (English-Marathi) Streamlit dashboardsuitable for non-technical agricultural department staff.

4. Dataset Description

This project uses a publicly available, secondary dataset of Maharashtra district-wisecropproduction statistics, merged with district-level rainfall and temperature records. The dataset isgovernment-origin agricultural data and satisfies the assignment's requirement of a minimumof 500observations several times over.

Raw rows (before cleaning)	
Raw columns	8: district, year, crop, area (1000 ha), production (1000 t), yield(kg/ha rainfall (mm), avg. temperature (°C)
Districts (raw)	26
Crops	14
Year range (raw)	1997 – 2017
Duplicate rows removed	7,588

Final model-ready rows	4,684
Final columns	14 (8 raw + 6 engineered)
Districts (final)	23 (3 dropped -- insufficient 3-year prior history)
Year range (final)	1999 – 2017

Target variable	risk_level -- categorical, 3 classes (Low / M
Granularity	district × crop × year

The district × crop × year granularity (rather than district × year alone) is a deliberate designchoice: distress risk genuinely differs by crop within the same district and year -- cotton and sugarcaneinthesame drought do not experience the same degree of stress -- and this granularity lets the dashboardgive crop-specific advice.

Variable Description

district, crop, year	Identifiers for the dis
area_1000ha	Area sown, in thousa
rainfall_mm	Total s
avg_temp_c	Average temperature
rainfall_deviation_pct	This ye
prev_year_yield	Yield of the same dist immediatelyprecedir
prior_3yr_avg_yield	Average yiel
prior_3yr_yield_volatility	Standard deviation of
yield_kg_per_ha	Actual yiel
distress_score	Composite of yield sh

	normalised
risk_level	Target label -- L

prior 3

5. Exploratory Data Analysis

Exploratory analysis was carried out on both the raw dataset (prior to cleaning) and the final model- ready dataset, to understand missingness, distribution shape, and the relationships betweencandidate predictors and the target.

Missing Values

Rainfall and temperature fields are missing for 18.45% of raw rows (2,800 of 15,176). Critically, thisisnot scattered randomly across years -- rainfall and temperature are entirely missing for theyears2010 and 2011 (all 1,456 rows for those two years), a genuine two-year gap in the source recordsrather than isolated missing entries.

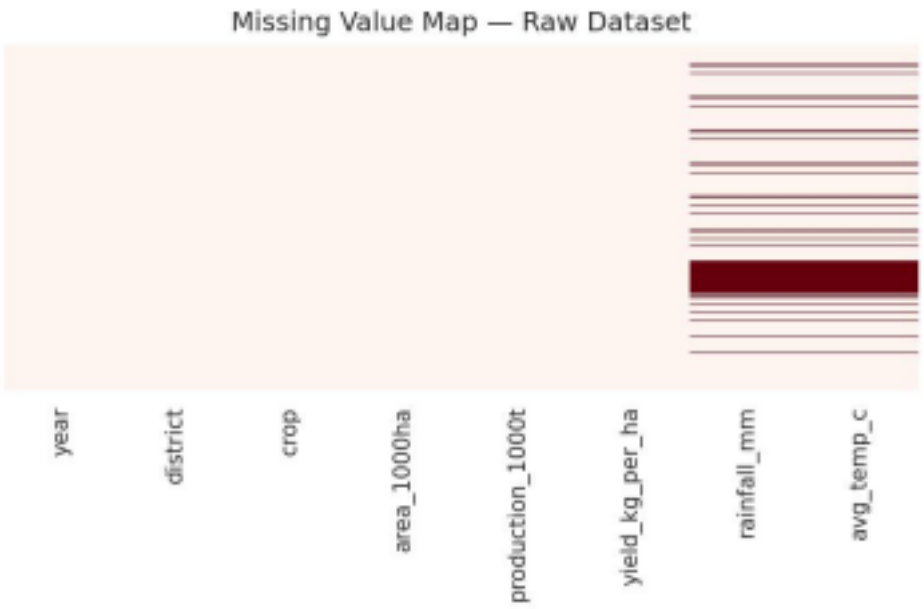


Figure 1. Missing value map -- raw dataset. The solid red bands correspond to the 2010-2011 rainfall/temperature gap.

Distributions

Rainfall is right-skewed with most district-years falling between 400-900mm. Yield is strongly right- skewed (skewness ≈ 3.45), driven by high-yield crops such as sugarcane sitting on a fundamentallydifferent scale from crops such as cotton or jowar.

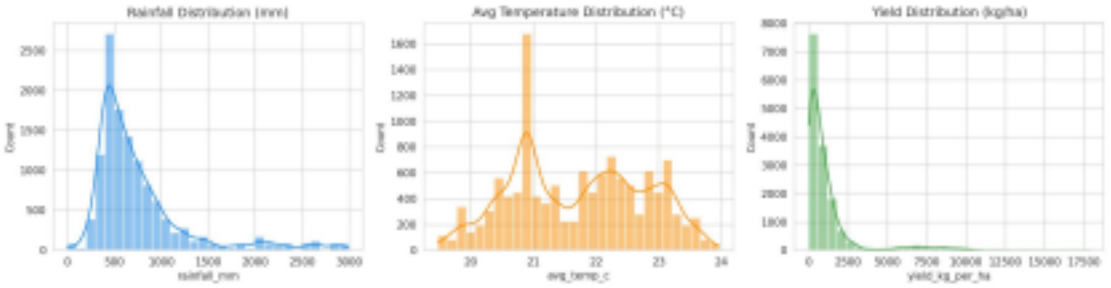


Figure 2. Distributions of rainfall, average temperature, and yield (raw dataset).

Trends Over Time

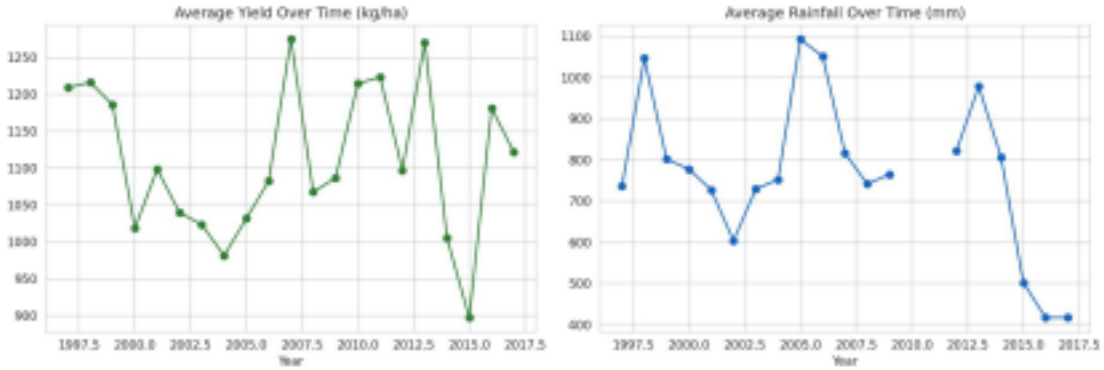


Figure 3. State-average yield and rainfall trends, 1997–2017. Visible dips coincide with known drought years.

Correlation With Distress

Correlating each numeric predictor against distress_score in the final dataset producedaclearranking: rainfall_deviation_pct is by far the strongest correlate (-0.51), followed by rainfall_mm(- 0.26) and avg_temp_c (+0.13); all yield-history features correlate weakly (≤ 0.09). A secondaryfinding was that prev_year_yield and prior_3yr_avg_yield correlate with each other at 0.98 -- near- perfect multicollinearity between two predictors, which is one reason a tree-based model (robust tocollinear inputs) was preferred over a purely linear one for the main model.

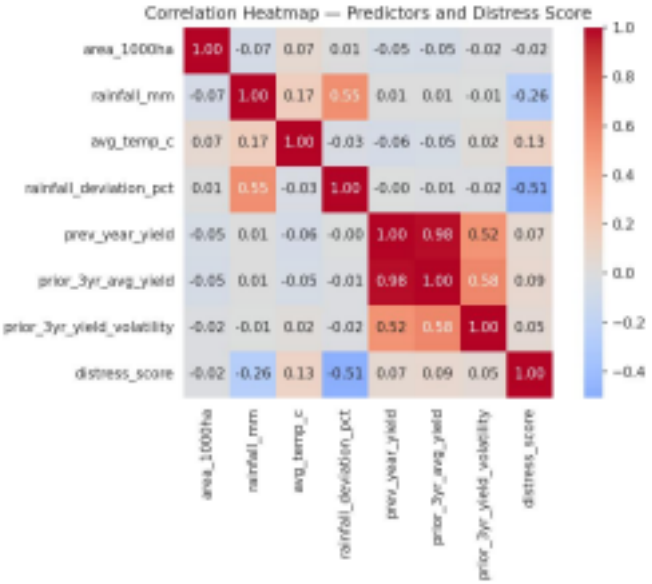


Figure 4. Correlation heatmap of predictors and distress_score.

Target Balance

The target variable is close to balanced by construction (Low: 1,546, Medium: 1,592, High: 1,546records), since risk_level is built from tertiles of distress_score. This makes plain accuracyareasonably fair headline metric, though macro-averaged F1 is also reported throughout as asafeguard.

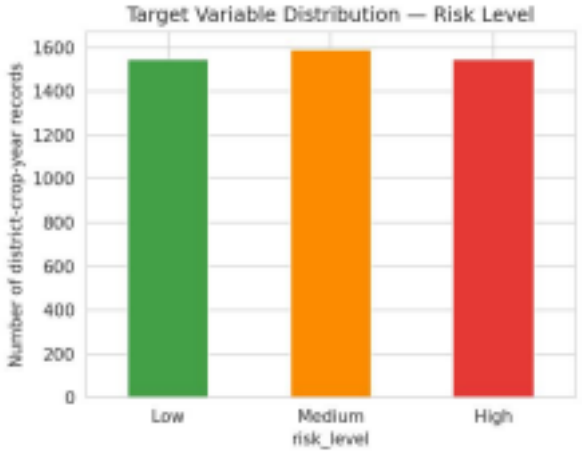


Figure 5. Target class distribution.

Highest-Risk Districts

Aggregating average distress_score by district surfaces the districts with the highest historical average risk over the full period -- a useful real-world sanity check, since these districts broadlycorrespond to Maharashtra's known drought-prone regions.

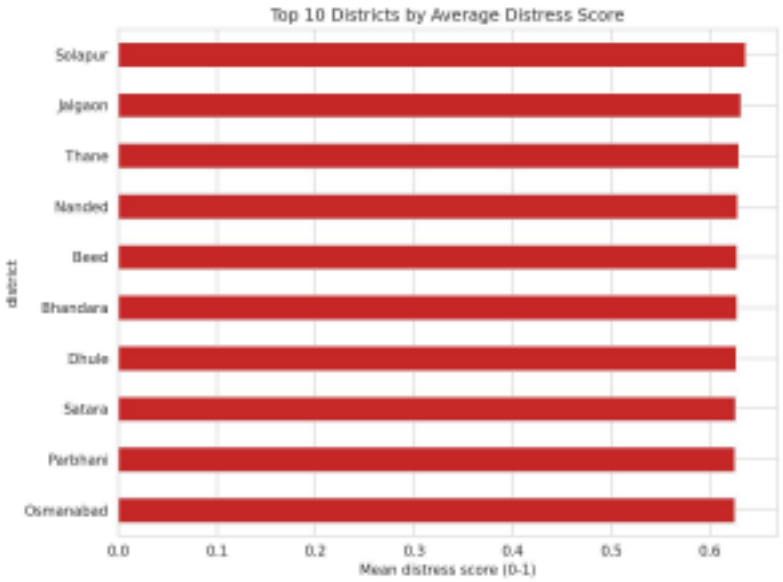


Figure 6. Top 10 districts by average distress score.

6. Data Pre-processing

Duplicate and Missing Value Treatment

7,588 exact duplicate district-year-crop rows were removed from the raw 15,176 rows beforefurther cleaning. Rows with missing rainfall or temperature were then dropped rather than imputed, since imputing weather data risks fabricating the exact drought/surplus signal the model exists todetect.

Outlier Detection (IQR Method)

The interquartile range (IQR) method was applied to the key raw numeric variables. The tablebelowreports the count and share of statistical outliers found in each.

area_1000ha	1,518	10.00%	-88.10	147.90
production_1000t	1,828	12.05%	-81.36	136.21
yield_kg_per_ha	1,148	7.56%	-1223.08	2653.85
rainfall_mm	980	7.92%	-257.53	1541.72
avg_temp_c	0	0.00%	18.43	25.01

These outliers were deliberately not removed. In this domain, an extreme yield or rainfall valueisfrequently a genuine drought or bumper-crop year -- precisely the distress signal this project existsto detect -- so deleting such rows would remove the phenomenon of interest rather than noise.

Feature Engineering

The following features were engineered, all computed using only prior years' data (via a one-periodlag per district-crop group) to guarantee no future information leaks into a prediction:

- rainfall_deviation_pct -- this year's rainfall vs. that district's own historical average (district- relative, not a blanket state average).
- prev_year_yield, prior_3yr_avg_yield, prior_3yr_yield_volatility -- lagged yield statistics per district-crop group.
- distress_score -- composite of yield shortfall vs. 3-year prior average (weighted 65%) andrainfall deviation (weighted 35%), min-max normalised to [0, 1].
- risk_level -- target label, built by tertile-splitting distress_score into Low / Medium/ High.

Leakage Safeguard

The current year's actual yield is used only to build the label, never as a predictor. This wasadeliberate design correction: an earlier version of this pipeline included the current year's yieldindirectly among the predictors and produced an unrealistic, artificially inflated accuracy closeto99% -- a textbook symptom of target leakage. Restricting all yield-based features to strictly prioryears, as described above, fixed this and produced the more modest, genuinely predictive resultsreported in Section 8.

Encoding and Scaling

The target label was encoded via LabelEncoder (Low / Medium / High →0 / 1 / 2). Predictorcategoricals -- district and crop -- were not one-hot encoded directly; their influence enters themodel indirectly through group-specific lagged yield statistics instead, avoiding a 23×14 dummy- variable expansion. This is discussed further as a limitation in Section 11. Feature scaling(StandardScaler) was applied only for Logistic Regression, which is scale-sensitive; it wasintentionally not applied for XGBoost, since tree-based splits are scale-invariant.

7. Model Development

Two models were developed and compared on identical data and splits, in line with the assignment'srequirement to benchmark against a suitable baseline.

Baseline: Logistic Regression

A multinomial Logistic Regression was used as the baseline model -- a standard, interpretable, andwidely accepted comparison point for a classification task of this kind.

Main Model: XGBoost

XGBoost, a gradient-boosted tree ensemble, was selected as the main model. This choice is justifiedby the nature of the problem: the relationship between rainfall deviation, temperature, andyieldhistory is plausibly non-linear and interactive -- for instance, a rainfall deficit likely matters morewhen sown area is large -- a relationship a linear model cannot represent but a tree ensemblecan. XGBoost is also inherently robust to the multicollinearity identified in Section 5 (prev_year_yieldvs. prior_3yr_avg_yield, r = 0.98), which would otherwise inflate coefficient variance in a linear model.

Final Feature Set

Seven predictors were used for both models: area_1000ha, rainfall_mm, avg_temp_c, rainfall_deviation_pct, prev_year_yield, prior_3yr_avg_yield, and prior_3yr_yield_volatility.

Hyperparameters and Training Procedure

n_estimators	200
max_depth	4
learning_rate	0.08
subsample	0.8
colsample_bytree	0.8
objective	multi:softprob
num_class	3

Data was split 80/20 into training and test sets using stratified sampling (random_state = 42forreproducibility), ensuring all three classes are proportionally represented in both splits.

8. Model Validation

Model performance was validated using both a held-out stratified test split and 5-fold stratifiedcross-validation, to check that results are not an artifact of a single lucky split.

Test Set Results

Logistic Regression(baseline)	57.5%		0.57	0.570	0.761

XGBoost (main	model) 63.9%		0.64	0.638	0.816

5-Fold Stratified Cross-Validation (F1-macro)

XGBoost	0.632	± 0.024
Logistic Regression	0.580	± 0.028

The cross-validation mean for XGBoost (0.632) sits close to its single-split test score (0.638), indicating the result is stable and not a lucky split.

Per-Class Performance (XGBoost, Test Set)

High	0.70	0.71	0.71	309
Low	0.65	0.70	0.68	309
Medium	0.55	0.51	0.53	319

Confusion Matrix (XGBoost)

High	220	32	57
Low	19	216	74
Medium	74	82	163

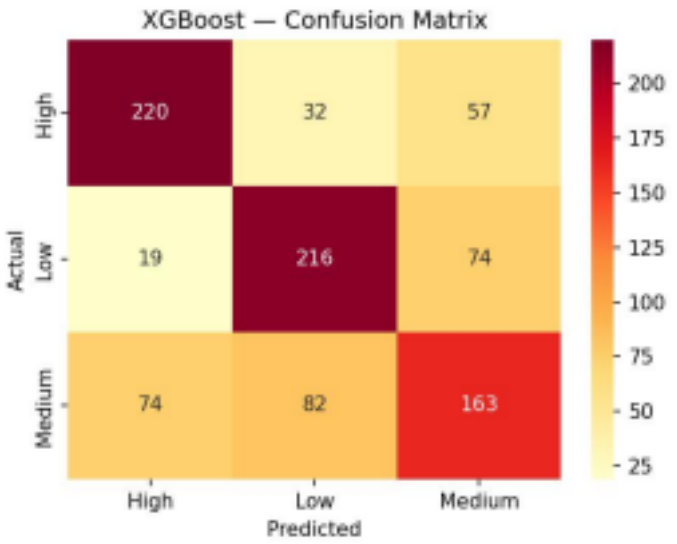


Figure 7. Confusion matrix, XGBoost, test set.

Feature Importance

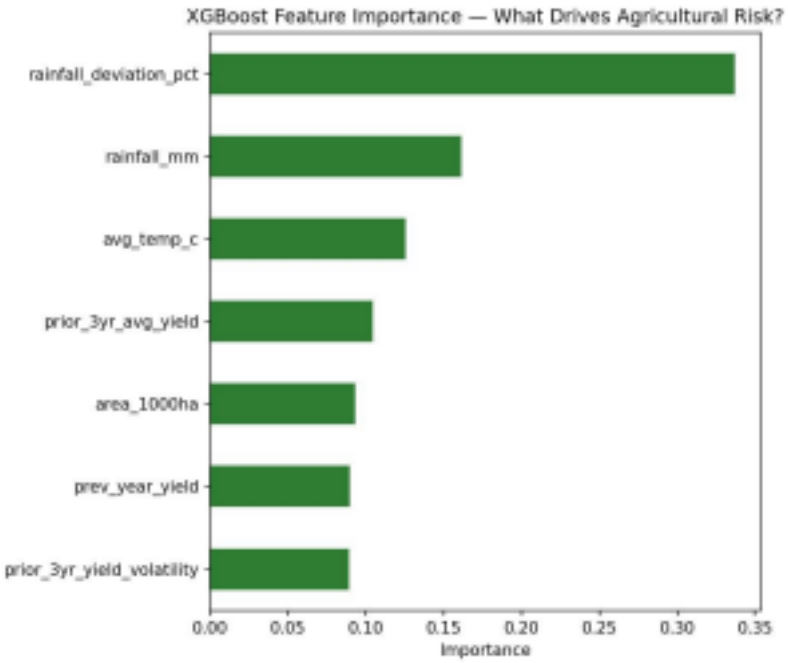


Figure 8. XGBoost feature importance ranking.

9. Results and Discussion

XGBoost outperformed the Logistic Regression baseline on every metric measured: +6.4 percentagepoints in accuracy, +0.068 in macro-F1, and +0.055 in ROC-AUC. This confirms that the non-linear, interaction-aware structure of a gradient-boosted tree ensemble adds genuine value over a linearbaseline for this problem, rather than the improvement being marginal or within noise.

The Medium risk class is consistently the hardest to classify (F1 = 0.53, versus 0.71 and 0.68 for Highand Low respectively). This is an interpretable, structural finding rather than a modelling flaw: riskisfundamentally a continuous underlying quantity that has been artificially discretised into threebins, so the Medium class sits between two comparatively easier extremes and absorbs most of theboundary confusion. The confusion matrix in Section 8 confirms this directly -- most Mediummisclassifications land in the adjacent High or Low classes, rarely at the opposite extreme.

Feature importance rankings from the trained XGBoost model closely mirror the correlationanalysisperformed during EDA (Section 5): rainfall_deviation_pct is the dominant feature (importance0.337), roughly twice the next-ranked feature (rainfall_mm, 0.161), followed by avg_temp_c (0.126). Yield- history features rank lowest (0.089-0.104). This internal consistency between an independent correlation analysis and the trained model's own learned importances is a meaningful validationsignal: the model is not picking up on spurious patterns, but rather amplifying the same rainfall- driven signal identified independently in the data.

This finding also directly supports the framing given in Section 1: rainfall deviation froma district'sown historical norm, not raw yield history, is this system's primary early-warning signal -- consistent with external research linking rainfall deficits to elevated farmer distress in Maharashtra.

10. Conclusion

This project built a leakage-free, government-data-based classification pipeline that separatesMaharashtra district-crop agricultural distress into three risk categories with 63.9%accuracyand0.638 macro-F1, meaningfully outperforming a linear baseline (57.5% accuracy, 0.570 macro-F1). Rainfall deviation from a district's own historical norm was identified, independently, as boththestrongest raw correlate of distress and the most important learned feature in the trained model -- the single strongest early-warning signal available in this dataset. The resulting model has beendeployed as an accessible, bilingual (English-Marathi) Streamlit dashboard, making the tool usablebynon-technical agricultural department staff, not only by data scientists.

11. Limitations

- Rainfall and temperature data has a genuine two-year gap (2010-2011) in the source records; those years are excluded entirely rather than imputed.
- Market price and financial loss data were not available at district level for this timeframe, sothe model uses yield/rainfall-based distress only -- the 'financial loss' dimension fromtheoriginal project concept is not modelled.
- district and crop are not directly encoded as model features; their influence enters only through group-specific historical statistics, which may lose some crop- or district-specific nuance a direct encoding could capture.
- Government district-level yield statistics carry a 2-3 year publication lag, which is why verified data could not be extended past 2017 for this project's timeframe.
- The Medium-risk class is inherently harder to classify than High/Low, a consequence of discretising a continuous risk score into three bins (Section 9).
- No satellite or image-based signal (vegetation health, drought imagery) is used -- the project is tabular-data-only.

12. Future Scope

- Incorporate satellite imagery (e.g., NDVI vegetation-health indices from Sentinel-2, freelyavailable via Google Earth Engine) as a genuinely complementary second signal -- a natural extension for a future CNN-based component.
- Extend the dataset forward as verified government data becomes available in later years, and test whether a model trained on 1999-2017 generalises to more recent years (a temporal generalisation check).
- Explicitly encode district and crop as categorical features (one-hot or target encoding) andmeasure whether this improves on the current implicit approach.
- Incorporate market price and financial loss data if a reliable source is found, restoring thefinancial-loss dimension from the original project concept.
- Add SHAP-based explainability for individual predictions, not just global feature importance, to strengthen the dashboard's 'explain why' capability.

13. References

1. Maharashtra district-wise crop production, rainfall, and temperature statistics -- secondary, government-origin agricultural dataset (see project GitHub repository for full source documentation).

2. National Crime Records Bureau (NCRB), "Accidental Deaths & Suicides in India" annual reports(2022-2023 editions) -- farmer/farm-labourer suicide statistics by state.

3. Bharadwaj, R., Karthikeyan, N., and Deulgaonkar, I., research on rainfall deficit and farmer suicidecorrelation across Indian states (2014-15 to 2020-21), as reported in Down To Earth, "Onefarmer/farm labourer dies by suicide every hour in India: NCRB data" (2023).

4. scikit-learn documentation -- https://scikit-learn.org

5. XGBoost documentation -- https://xgboost.readthedocs.io

6. Streamlit documentation -- https://docs.streamlit.io

7. Project GitHub repository -- https://github.com/ambredurva1113-byte/KrishiSahayak

14. Appendix

A. Code and Repository

Full source code, data files, trained model artifacts, and EDA outputs are available at:

https://github.com/ambredurva1113-byte/KrishiSahayak

B. Deployed Application

The dashboard is deployed on Streamlit Community Cloud and is publicly accessible. Seethedeployed application URL noted in the project README.

C. Dashboard Screenshot

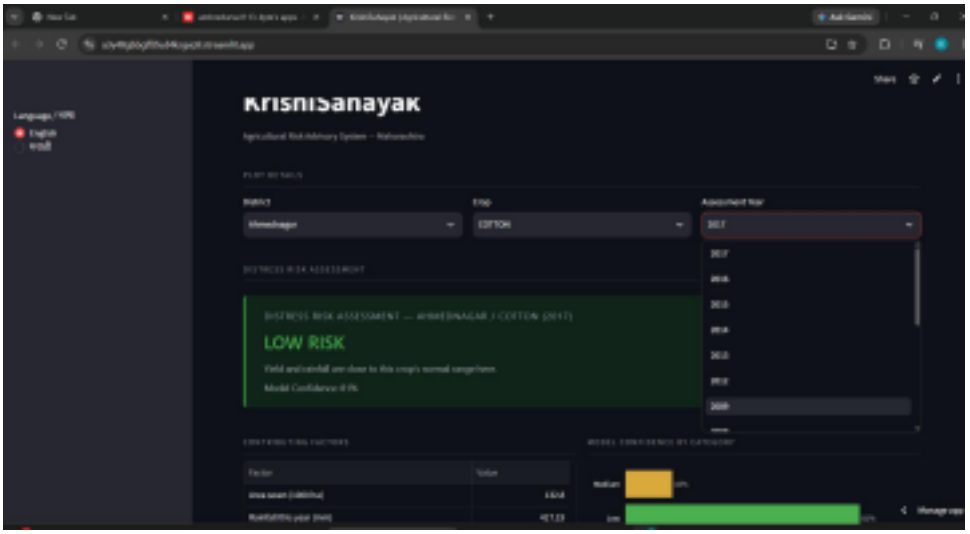


Figure 9. KrishiSahayak dashboard -- district/crop/year risk assessment view, showing risk level, contributingfactors, and model confidence.